

Predicting Benchmark Scores from Unpaired Model Evaluations

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Abstract

Evaluating a generative model on all available benchmarks is often prohibitively expensive. As such, models are often evaluated on different subsets of tasks and/or different subsets of queries from a given task, and the resulting (model $\times$ task) score matrix is sparse and noisy. In this paper, we introduce the **Product Kernel Perspective Space (PKPS)**—a collection of low-dimensional representations of models based on their embedded responses—that generalizes existing benchmark-prediction methods to the sparse and noisy evaluation regime by comparing responses to similar, rather than identical, queries. Theoretically, we show that PKPS more effectively uses information from unpaired queries for a given task. Empirically, we show that PKPS outperforms sample means, item-response theory, and paired methods where evaluation is sparse, noisy, or unpaired, and that it is a complementary signal to score-based methods.

1 Introduction

Generative models are evaluated on an ever-growing collection of benchmarks spanning knowledge, reasoning, code, translation, and safety [Liang et al., 2023]. Evaluating a model on all of them is often prohibitively expensive. As a result, models are evaluated on different subsets of tasks and on different subsets of queries within a task: a recent compilation of 84 frontier models on 133 benchmarks finds only 23% of the score matrix filled [Zeng and Papailiopoulos, 2026]. This sparsity is not an artifact of curation; it is a consequence of how evaluations are collected. Arena-style platforms score each model on the conversations its users generate [Chiang et al., 2024]; contamination-aware benchmarks refresh their items, so models evaluated in different months answer different questions [White et al., 2025]; and cost-driven subsampling gives each evaluation run its own item subset [Perlitz et al., 2024]. The resulting (model $\times$ task) score matrix is sparse, its entries are noisy averages over few queries, and the responses beneath it are *unpaired*: two models rarely answer the same queries. Predicting the missing and true entries of this matrix would let practitioners reuse cached evaluations instead of paying to re-run them [Polo et al., 2024, Vivek et al., 2024, Zeng and Papailiopoulos, 2026].

Two lines of work aim to reduce this cost by predicting scores rather than computing them. Efficient benchmarking extrapolates a task’s full score from a small scored subset of its items, classically by item-response theory [Polo et al., 2024]; matrix completion imputes cells that were never run, from low-rank structure across the matrix [Zeng and Papailiopoulos, 2026]. These are two ends of the same problem: every cell of the score matrix is observed with some number of queries, possibly zero, and predicting its true score amounts to denoising when that number is small and to completion when it is zero. A single representation of each model that is accurate enough to predict any cell would serve both ends.

Neither line of work has such a representation: both use only each cell’s aggregate score, and the responses that produced those scores carry more information than the scores themselves. The Data Kernel Perspective Space (DKPS) [Helm et al., 2026] uses this information: it embeds each model from its responses via classical multidimensional scaling and regresses benchmark scores in the resulting space. The DKPS distance between two models, however, is defined only over responses to shared queries—a *paired* design—and is undefined when models were evaluated on different queries. To our knowledge, no existing method predicts the score matrix from evaluation data as it is typically available: sparse, noisy, and unpaired.

Table 1: MAE of benchmark prediction methods on both suites (16 seeds; best per column bold; setup in Section 4.1): query efficiency at budgets $m \in \{1, 8\}$ queries per cell, and completion of missing cells at the small-cohort, full-cohort ($p_{\text{task}}=p_{\text{query}}=0.5$), and sparse-coverage (sp.: $p_{\text{task}}=0.2$, $n=10$) operating points. “—”: not applicable.

Method	HELM suite (18 tasks, 93 models)					EEE suite (16 tasks, 45 models)				
	Query-eff.		Completion			Query-eff.		Completion		
	($m=1$)	($m=8$)	($n=10$)	($n=93$)	(sp.)	($m=1$)	($m=8$)	($n=10$)	($n=45$)	(sp.)
Sample mean	0.292	0.092	—	—	—	0.224	0.081	—	—	—
IRT	0.348	0.343	—	—	—	0.165	0.139	—	—	—
LRMC	—	—	0.139	0.105	0.208	—	—	0.109	0.096	0.144
DKPS	0.168	0.109	—	—	—	0.116	0.077	—	—	—
PKPS	0.136	0.108	0.120	0.103	0.171	0.076	0.058	0.082	0.070	0.108
Ensemble	0.132	0.077	0.120	0.099	0.171	0.084	0.052	0.082	0.069	0.108

In this paper, we introduce the **Product Kernel Perspective Space (PKPS)**, a collection of low-dimensional representations of models based on their embedded responses. PKPS compares responses to *similar*—rather than only identical—queries, so every unpaired response contributes and paired DKPS is recovered as a special case (Section 3). Theoretically, we show that the query budget PKPS requires to reach a target accuracy does not depend on the fraction of queries shared across models (Theorem 1). Empirically, we demonstrate on two heterogeneous suites—93 models on HELM tasks and 45 modern models from the Every Eval Ever datastore [EvalEval Coalition, 2026]—that PKPS achieves lower error than sample means, item-response theory, and DKPS in sparse, noisy, and unpaired regimes, matching the accuracy of sample means with an $8\times$ smaller query budget, and that ensembling it with score-based methods reduces the error of low-rank matrix completion by up to 28% (Section 4).

1.1 Problem setting and notation

A *model* is a random mapping $f: \mathcal{Q} \rightarrow \mathcal{X}$ from a query space to a response space; querying f at q returns a draw from the distribution $f(q)$. We observe n models $f_1, \dots, f_n$, each run on its own *query set* $Q_i \subseteq \mathcal{Q}$; the sets may overlap arbitrarily, and the paired design $Q_1 = \dots = Q_n$ is a special case. Two fixed embedding functions map raw objects to vectors: a *response* embedding $\text{embed}_R: \mathcal{X} \rightarrow \mathbb{R}^p$ and a *query* embedding $\text{embed}_Q: \mathcal{Q} \rightarrow \mathbb{R}^{p'}$. For $q_j \in Q_i$ we write $x_{ij} = \frac{1}{r} \sum_{k=1}^r \text{embed}_R(f_i(q_j)^{(k)})$ for the average of the r sampled response embeddings and $u_j = \text{embed}_Q(q_j)$ for the embedded query.

Queries are grouped into T tasks. Model i ’s *true score* on task t , y_{it} , is its mean score over the task’s full query set; running the model on $m_{it} \geq 0$ of those queries yields an observed *sample score* whose noise scales as $1/\sqrt{m_{it}}$. The goal is to predict y_{it} for every cell of the (model $\times$ task) matrix: cells observed with few queries ($m_{it} > 0$; *denoising*, the efficient-benchmarking end) and cells never run ($m_{it} = 0$; *completion*). We write y_i when the benchmark has a single score. Two properties of the design recur throughout: *coverage*, the fraction of cells with $m_{it} > 0$, and *pairing*, the fraction ρ of a cell’s queries shared across models; real evaluations sit near $\rho \approx 0$.

2 Related work

Low-dimensional representations of models. Some approaches construct representations by comparing internal activations across models [Duderstadt et al., 2023, Huh et al., 2024, Horwitz et al., 2025] or by comparing model weights directly [Chen et al., 2025]. Evaluation frameworks, however, typically permit only black-box access: they observe responses, not internals. The Data Kernel Perspective Space (DKPS), first introduced in the context of monitoring multi-agent systems [Helm et al., 2024], addresses this setting by mapping a collection of models into a low-dimensional Euclidean space via classical multidimensional scaling of average embedded responses [Torgerson, 1952]. Prior work established consistency and concentration of the representations [Acharyya et al.,

2024, 2025], supervised inference thereon [Helm et al., 2025], and query efficiency [Helm et al., 2026]. These constructions compare responses to shared queries, a paired design. PKPS replaces the identity matching with a product kernel so that models evaluated on different queries remain comparable.

Benchmark prediction. A second line of work predicts unobserved (model, task) scores so that not every evaluation must be run; efficient benchmarking and matrix completion are its two ends. Polo et al. [2024] use item-response theory to construct small item subsets whose scores predict full performance [see also Rasch, 1960, Kipnis et al., 2025]; Vivek et al. [2024] cluster items by cross-model agreement and score only the cluster centers; Bean et al. [2025] select items using cognitive-scale embeddings; Li et al. [2024] select items adaptively via reinforcement learning; and Perlitz et al. [2024] analyze the cost–reliability tradeoff of benchmark designs. Relatedly, Ilyas et al. [2022] predict a model’s outputs as a function of its training data, Prabhu et al. [2024] reuse past evaluations in lifelong benchmarks, and Helm et al. [2026] predict held-out scores from cached responses. On the completion end, Zeng and Papailiopoulos [2026] observe that the (model $\times$ task) score matrix is approximately rank-two and complete it in logit space [see also Candès and Recht, 2009, Mazumder et al., 2010, Koren et al., 2009]. These methods use only each cell’s aggregate score. PKPS unifies the two ends—a cell with $m \geq 0$ queries spans efficient benchmarking ($m > 0$) and completion ($m = 0$)—and adds a response-level signal that can be ensembled with matrix completion.

3 The Product Kernel Perspective Space

Models are embedded by classical MDS of pairwise affinities. Each method summarizes the n models of Section 1.1 in an $n \times n$ affinity matrix A (defined below). The squared distance between two models is $D^2(a, b) = A_{aa} + A_{bb} - 2A_{ab}$ (clamped at 0), and their d -dimensional *representations* are the classical-multidimensional-scaling minimizer

$$(\hat{\psi}_1, \dots, \hat{\psi}_n) = \arg \min_{z_1, \dots, z_n \in \mathbb{R}^d} \sum_{i, k=1}^n (\|z_i - z_k\| - D(i, k))^2, \quad (1)$$

so model i is the point $\hat{\psi}_i \in \mathbb{R}^d$, its *perspective*. Paired DKPS and PKPS differ only in the affinity A .

Paired DKPS compares responses to identical queries only. With all n models answering the same m queries ($Q_1 = \dots = Q_n$), DKPS uses the identity-matched affinity $A_{ab} = \frac{1}{m} \sum_j k_R(x_{aj}, x_{bj})$: response j of model a is paired only with response j of model b , so the design must be paired.

PKPS weights every response comparison by query similarity. PKPS replaces identity matching with a query kernel k_Q :

$$A_{ab} = \frac{\sum_{j \in Q_a} \sum_{l \in Q_b} k_Q(q_j, q_l) k_R(x_{aj}, x_{bl})}{\sum_{j \in Q_a} \sum_{l \in Q_b} k_Q(q_j, q_l)}, \quad (2)$$

where Q_a is the query set of model a . Each term is normalized by its own query-kernel mass, so models that answered different numbers of (or entirely different) queries are still comparable; responses to *similar* queries—even across tasks—contribute and unpaired evaluations are used (Figure 1). Paired DKPS is the special case $k_Q = \delta$, where only $j = l$ survives.

An RBF query kernel; a linear or RBF response kernel. The response kernel k_R compares embedded responses, the query kernel k_Q compares embedded queries. Paired DKPS uses a linear

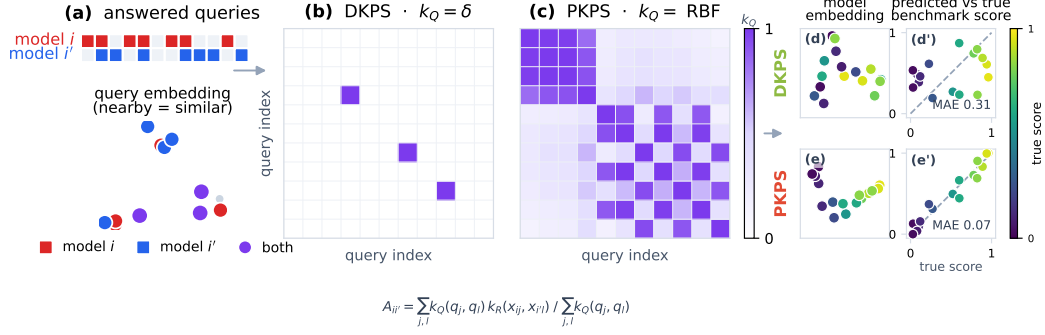


Figure 1: PKPS uses every response under unpaired evaluation; DKPS uses only the shared ones. **(a)** Two models, i (red) and i' (blue), answer their own subsets of a query pool (blocks: who answered what; dots: queries in embedding space, nearby = similar). **(b)** DKPS’s identity kernel keeps only the diagonal cells where both models answered the *same* query—almost every response is discarded. **(c)** An RBF k_Q is a dense similarity matrix: responses to *similar* queries bridge the two models. Both panels index the pooled queries in one shared order; the weights feed one entry of the model affinity (Eq. 2). **(d–e’)** A toy benchmark-prediction example: restricted to the rare shared queries, DKPS’s embedding has no clean score axis and predicts poorly (MAE 0.31); PKPS recovers a smooth score gradient and predicts on the diagonal (MAE 0.07).

response kernel and the identity query kernel $k_Q = \delta$. We use an RBF query kernel over the embedded queries, $k_Q(q_j, q_l) = \exp(-\|u_j - u_l\|^2 / 2\sigma^2)$, with bandwidth σ chosen per run (below); k_R is RBF in the synthetic study and linear on real data, where linearity makes the kernel factorize across response domains (Section 4.1). Since Eq. 2 normalizes each pair by its own kernel mass, A need not be positive semidefinite; classical MDS embeds the doubly-centered matrix by its top- d nonnegative eigenvalues, as is standard. A_{ab} costs $|Q_a||Q_b|$ kernel terms per pair, and one response-kernel pass serves the whole bandwidth grid.

3.1 Theoretical properties of the PKPS

PKPS is a strict generalization of DKPS. With $k_Q = \delta$ only equal queries lying in both query sets survive the double sum of Eq. 2, so it reduces to paired DKPS over the shared queries, $A_{ab} = \frac{1}{|Q_a \cap Q_b|} \sum_{q \in Q_a \cap Q_b} k_R(x_{aq}, x_{bq})$; since the RBF kernel converges pointwise to δ as $\sigma \rightarrow 0$, DKPS is also the small-bandwidth limit. Conversely, for any strictly positive k_Q the normalizer is positive for any nonempty query sets, so A_{ab} is defined even on disjoint sets, where paired DKPS is undefined; between the limits, σ trades exact matching for bridging.

3.2 Inference in the PKPS

Inference is a two-step statistical problem, exactly as in the DKPS [Helm et al., 2026]: embed each model into the perspective space by the scaling of Eq. 1, then regress its benchmark score on the resulting coordinate. On reference models with known scores $\{(f_i, y_i)\}$ we fit $\hat{h} : \mathbb{R}^d \rightarrow \mathcal{Y}$ on the pairs $\{(\hat{\psi}_i, y_i)\}$ and predict $\hat{y}(f) = \hat{h}(\hat{\psi}(f))$. The free choices are the bandwidth σ and the regressor $\hat{h}$.

Leak-free cross-validation chooses the bandwidth. The optimal σ depends on how paired the data is: small when query overlap is high, large when it is low. We choose it per run by cross-validation: over a median-scaled grid $\sigma \in \{0.03, \dots, 3\} \cdot \text{med}$, select the σ whose perspective best predicts the *observed* scores under leave-one-model-out k -NN. The criterion never sees the held-out

scores we report. And because the grid reaches near-delta, cross-validation *recovers* *DKPS* when overlap is high and bridges only when it must.

The estimation pipeline shares the matrix-completion head. For each replicate we (i) PCA-reduce response and query embeddings on the *observed* rows only (dimension by the second Zhu–Ghodsí elbow); (ii) form D from Eq. 2; (iii) embed models by classical MDS; (iv) regress scores on the embedding, fit leave-one-family-out. The score head—logit transform, per-task standardization, two-way (model + task) bias—is identical to the low-rank matrix-completion (LRMC) baseline’s; the two methods differ only in the factor, k -NN on the perspective versus a low-rank factorization, so comparisons isolate the representation. Matching the per-task standardization matters: without it the model offset is dominated by the highest-logit-variance tasks.

The ensemble folds in whichever score signal a cell has. The rule is precision weighting: trust each channel in proportion to its estimated reliability. An *observed* cell’s score signal is its own sample mean, shrunk toward the PKPS estimate by the weight $\sigma_q^2/(\sigma_q^2 + e_{PKPS}^2)$ —sample noise variance over total disagreement with PKPS—so the sample dominates as queries grow and the embedding as noise grows. A *missing* cell’s score signal is LRMC [Zeng and Papailiopoulos, 2026], blended with PKPS by a weight cross-validated on observed scores. Every weight is read off observed queries alone ($\hat{p}(1 - \hat{p})/m$; depth-weighted disagreement); none touches the held-out scores we report.

3.3 Query efficiency

DKPS requires query *sets* to overlap; PKPS requires only query *distributions* to overlap. This section makes that distinction precise by extending the machinery of Helm et al. [2026].

Definition 1 (Population PKPS). Let each model a ’s (query, response) pairs be drawn i.i.d. from a design distribution P_a . The *population affinity* is $A_{ab}^* = \mathbb{E}[k_Q(u, u') k_R(x_a, x_b)] / \mathbb{E}[k_Q(u, u')]$ with draws from P_a and P_b independent, the population squared distances are $D^{*2}(a, b) = A_{aa}^* + A_{bb}^* - 2A_{ab}^*$, and the *population perspectives* $\psi_1, \dots, \psi_n \in \mathbb{R}^d$ are the rank- d spectral embedding of the centered Gram matrix B^* of D^{*2} .

Assumption 1 (Score regularity). The benchmark score is bounded and γ -Lipschitz on the population perspective space.

In practice, Assumption 1 holds when models with similar response distributions attain similar scores—the premise of any response-based predictor, and visible as the smooth score gradient in Figure 1(e).

Assumption 2 (Model support). The model distribution places mass on every neighborhood of the population perspective space.

Assumption 2 ensures that a target model has arbitrarily near reference neighbors once the cohort is large; in practice it suggests that a large, diverse reference population may be needed. Both conditions mirror Helm et al. [2026].

Assumption 3 (Kernel and geometry regularity). Kernels and embedded responses are bounded; the *cross-model query-kernel mass* is bounded below, $\kappa = \inf_{a,b} \mathbb{E}[k_Q(u, u')] > 0$; and B^* has rank d with spectral gap $\lambda_d > 0$.

Boundedness and the spectral gap are standard in spectral-embedding analyses. The substantive condition is $\kappa > 0$: it is the population analogue of the pairing fraction ρ , and for an RBF k_Q it holds whenever query *distributions* overlap, even when no individual query is shared. It fails only when query distributions are disjoint—models evaluated on domains the kernel cannot bridge—in which case no unpaired method should be expected to compare them.

Proposition 1 (Concentration of PKPS perspectives). *Under Assumption 3, with at least m queries per model, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$ there is an orthogonal W such that $\max_i \|\hat{\psi}_i - W\psi_i\| \leq C\kappa^{-2}\lambda_d^{-1/2}\sqrt{dn\sqrt{\log(4n^2/\delta)}/2m}$, where C depends only on the kernel bounds (proof in Appendix D).*

Definition 2 (Query efficiency; following Helm et al., 2026). Fix a per-model budget m and let h_n denote a score estimator using n scored reference models. h is *query-efficient relative to h' at budget m* if there exists N such that $\text{MSE}(h_n) \leq \text{MSE}(h'_n)$ with high probability for all $n \geq N$, and *query-efficient relative to h'* if this holds at every budget at which $\text{MSE}(h'_n)$ is bounded away from zero.

Theorem 1 (Query efficiency of PKPS). *Under Assumptions 1–3, for any $\varepsilon > 0$ there exist (n, m) , with m polynomial in $(1/\varepsilon, 1/\kappa, n)$ and independent of the pairing fraction ρ , such that nearest-neighbor regression in the PKPS attains mean squared error at most ε with high probability. That is, whenever the sample score’s error at that budget remains above ε —its MSE is σ_q^2/m for every n , where σ_q^2 is the per-query score variance—PKPS is query-efficient relative to the sample score, at every pairing fraction including $\rho = 0$.*

Proof sketch. Proposition 1 bounds the estimation error of every perspective. By Assumption 1 and the triangle inequality, the nearest-neighbor regression error splits into two estimation terms and the population nearest-neighbor distance, which Assumption 2 makes small for n large; choosing $m = \text{poly}(n, 1/\kappa, 1/\varepsilon)$ gives $\text{MSE} \leq \varepsilon$, and no step involves ρ . The sample score uses no reference models, so its error at budget m is fixed at σ_q^2/m ; attaining ε below it gives the relative statement. Full proof in Appendix D. $\square$

Applied to paired DKPS ($k_Q = \delta$), the same argument retains only the ρM_{ij} shared query pairs, so the DKPS bound’s effective budget is ρm and its guarantee degrades as $\rho \rightarrow 0$ while Theorem 1 does not. In the sense of Definition 2 we conjecture strict separation; Section 4.2 is its empirical counterpart.

Conjecture 1 (Strict gap). *Under Assumptions 1–3, PKPS is query-efficient relative to paired DKPS at every budget, with the MSE gap widening as $\rho \rightarrow 0$; the two coincide in the fully paired limit $\rho = 1$.*

4 Experiments

We evaluate PKPS on synthetic data, where every lever of the observation process can be swept, and on two heterogeneous benchmark suites. Throughout, the two prediction problems of Section 1.1 structure the experiments: denoising observed cells (query efficiency) and predicting missing ones (completion).

4.1 Evaluation setup

Synthetic data. We generate n models with latent ability vectors v_i and T tasks with latent vectors t_k , scores $v_i \cdot t_k + \varepsilon$, and queries drawn near each task center. Each model answers M_{ij} queries per task, of which $m_{i'j'}$ are shared with another model (overlap $\rho = m_{i'j'}/M_{ij}$); a task is observed with probability p_{task} and a query with probability p_{query} .

Benchmark suites. The first suite is 18 tasks across four HELM datasets with very different response types—MATH (7 math subjects), WMT-14 (5 translation directions), MedQA (medical multiple-choice), and LegalBench (5 legal classification subsets)—on the 93 models evaluated on

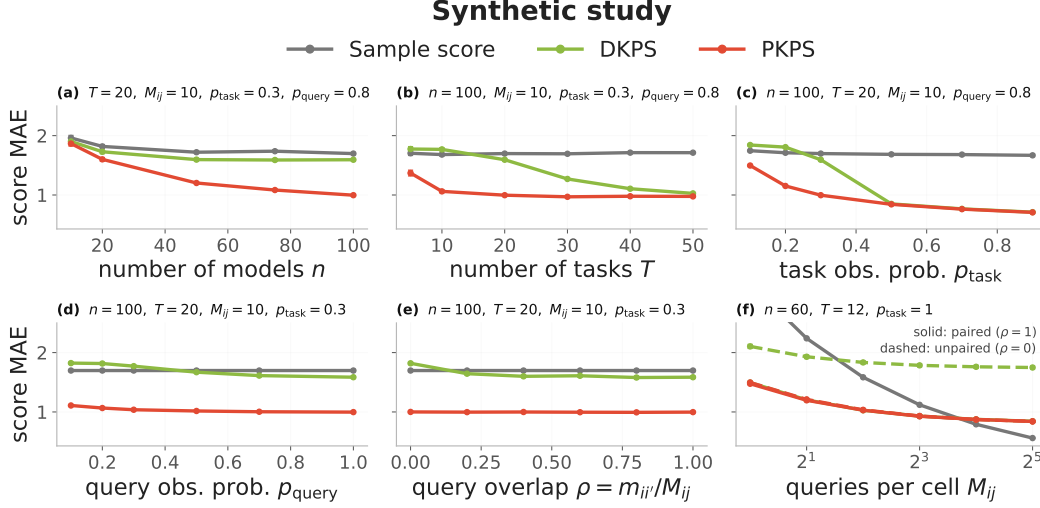


Figure 2: PKPS outperforms DKPS on every lever of the synthetic benchmark (score MAE, ± 1 SEM over 50 seeds). DKPS (identity query kernel, green) vs. PKPS (RBF, red), with the score-only sample baseline (gray); each panel sweeps one parameter with the rest fixed (shown in its title). **(a)–(e) complete held-out cells** from observed responses—cohort n , tasks T , task coverage p_{task} , query coverage p_{query} , and cross-model overlap $\rho = m_{i'j'}/M_{ij}$. PKPS (CV-selected bandwidth) beats DKPS throughout—by a wide margin even when mostly paired (a–c), since it bridges across tasks as well as within—and the gap is largest when unpaired or sparse (d, e), where in (e) DKPS collapses toward the floor as $\rho \rightarrow 0$ while PKPS stays flat. **(f) denoises observed cells**: each cell is measured with M_{ij} noisy queries and we predict its true score, at paired ($\rho=1$, solid) and unpaired ($\rho=0$, dashed). The raw sample falls $\propto 1/\sqrt{M_{ij}}$; smoothing over the perspective denoises well below it at small budgets—PKPS paired or unpaired, DKPS only paired—until the sample becomes precise at the full budget.

all four. MATH and WMT have rich free-text responses; MedQA and LegalBench responses are single tokens for which text embeddings are degenerate, so we encode each dataset’s responses natively (Gemini embeddings vs. one-hot answers) in disjoint blocks. With a linear k_R this makes cross-dataset response similarity exactly zero, and a single shared query embedding with a within-domain bandwidth keeps k_Q near zero across domains: the product kernel factorizes by domain, never comparing a proof to a legal label. The second suite, from the Every Eval Ever datastore [EvalEval Coalition, 2026], is 45 modern models on 16 tasks (MATH-MC levels 1–5, GSM-MC, GPQA-Diamond, and nine judging categories from JudgeBench and RewardBench 2), with per-cell query pools drawn independently per model ($\sim 10\%$ incidental overlap)—unpaired by construction (Appendix B).

Prediction methods. We compare six predictors. The *sample mean* is a cell’s own m -query average. *IRT* fits a one-parameter logistic model on the largest shared item block [Polo et al., 2024]. *Low-rank matrix completion* (LRMC) is the logit-space imputation of Zeng and Papailiopoulos [2026] (missing cells only). *DKPS* and *PKPS* regress scores on the model embeddings of Section 3, DKPS being the identity-query-kernel limit. The *ensemble* combines PKPS with whichever score signal a cell has (Section 3).

Measurement. Every experiment predicts held-out or true cell scores and reports MAE against the full-evaluation score, mean ± 1 SEM over 16 seeds (50 on synthetic data). Sweeps vary one observation lever at a time, holding the others fixed (grids in Appendix A); all hyperparameters, including the query bandwidth, are chosen per run by the leak-free cross-validation of Section 3.

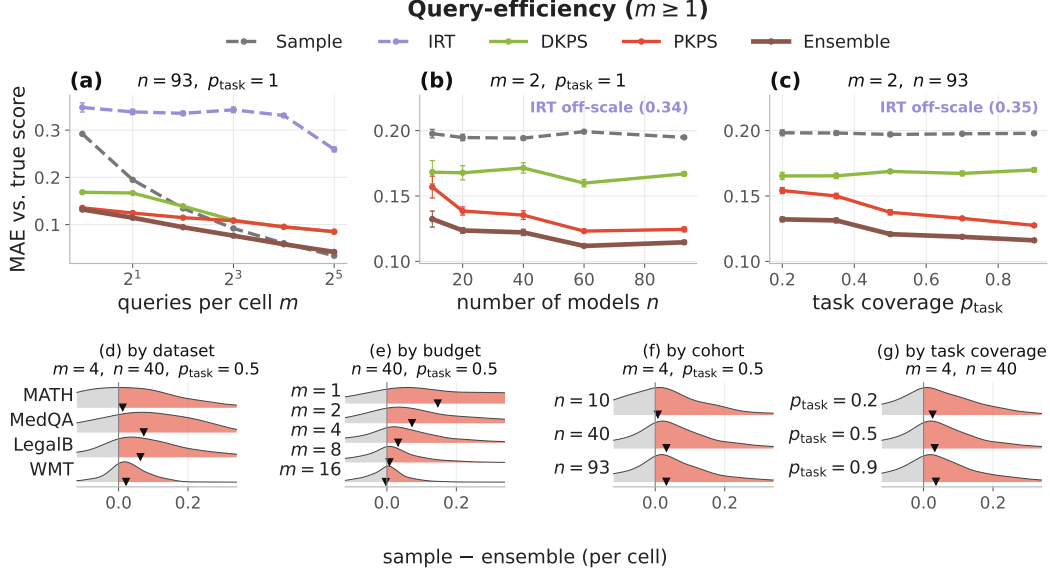


Figure 3: Regression in the PKPS provides query-efficient benchmark prediction relative to the sample score. (a–c) Each model answers m unpaired queries per *observed* cell; predict its true full-eval score (MAE, ± 1 SEM). PKPS (red) and the ensemble (brown) denoise far below the raw sample (grey) at small budgets and across cohort (b) and coverage (c); DKPS (green) trails on unpaired queries, and IRT (purple) is poorly identified without a shared item block (omitted from b, c; value annotated). (d–g) Conditional breakdown: per-cell $\Delta = \text{sample} - \text{ensemble}$ MAE ($\Delta > 0$, red = win) by dataset, budget, cohort, and coverage, other levers at their sweep median. Broad but tail-driven; the per-row mean ($\blacktriangledown$) is positive throughout.

4.2 Results

PKPS outperforms DKPS on every observation lever. We predict held-out (model, task) scores against the score-only sample baseline (Figure 2). PKPS beats DKPS on *every* lever, and both beat the sample floor: bridging similar queries—across tasks as well as within—supplies signal that exact matching cannot reach, so the margin is wide even when the data is mostly paired (a–c). The gap peaks where pairing is scarce: as p_{query} drops (d) DKPS degrades while PKPS stays flat, and sweeping ρ directly (e), DKPS collapses toward the floor as $\rho \rightarrow 0$ while PKPS is flat—it does not rely on pairing.

Panel (f) is the *denoising* form of the same problem: each cell is now *observed* with M_{ij} noisy queries (sample noise $\propto 1/\sqrt{M_{ij}}$) and we predict its true score. The raw sample mean falls like $1/\sqrt{M_{ij}}$; smoothing those noisy scores over the perspective denoises well below it. PKPS at one query matches the sample at roughly five, paired or unpaired; DKPS denoises only when paired (dashed). Only at large budgets is the raw sample precise enough to overtake the embedding’s residual bias. This is the empirical counterpart of Theorem 1 and Conjecture 1.

PKPS-based methods outperform the sample score at small query budgets. Each model answers m queries per cell, sampled independently—the real-data $\rho \rightarrow 0$ regime, on which IRT is poorly identified—and we predict the true full-eval score (Figure 3). The embedding captures response *style*, which is stable even when the per-cell sample mean is noisy, so it denoises cheap evaluations. At a small budget the embedding is far more accurate than the raw mean: at $m=1$, PKPS scores 0.136 and the ensemble 0.132 versus the sample’s 0.292; PKPS at one query matches the sample at four. As the budget grows the raw mean overtakes the embedding’s residual bias (0.034 vs. 0.085 at $m=32$); the ensemble tracks the better channel throughout. Panels (b, c) separate the embeddings: PKPS improves steadily with cohort and coverage (0.157 $\rightarrow$ 0.124, 0.154 $\rightarrow$ 0.128;

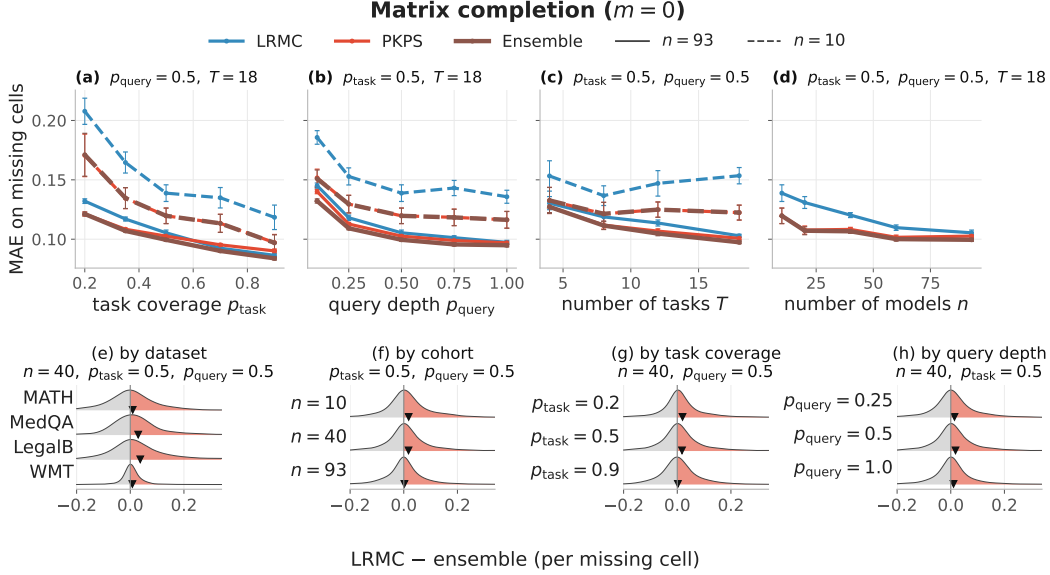


Figure 4: The PKPS embedding rescues the small-cohort regime where LPMC collapses for lack of rows, and ties it at full cohort. (a–d) Observe a fraction of (model, task) cells, predict the *missing* ones (MAE on missing cells, ± 1 SEM). Colour is method (LPMC blue, PKPS red, ensemble brown); line style is cohort (solid $n=93$, dashed $n=10$). (e–h) Conditional breakdown: per-missing-cell $\Delta = \text{LPMC} - \text{ensemble}$ MAE ($\Delta > 0$, red = win) by dataset, cohort, coverage, and query depth, other levers at their sweep median. Modest and tail-driven on this low-rank suite, but the per-row mean ($\blacktriangledown$) stays positive.

35–37% on the second suite) while DKPS, which cannot use the additional unpaired responses, stays flat.

The embedding completes missing cells where LPMC cannot. The same embedding also predicts cells that were *never run*. We observe a fraction p_{task} of cells (each at depth p_{query}) and predict the missing ones, against the LPMC baseline (the sample does not apply—a missing cell has no sample). Low-rank completion needs many rows: when models are scarce it fails while the embedding remains accurate—at $n=10$, LPMC is 0.139 versus PKPS’s 0.120 (Figure 4, dashed). At full cohort ($n=93$, solid) PKPS slightly edges LPMC (0.103 vs. 0.105) and the ensemble is best (0.099).

The per-cell wins are broad but tail-driven. The ensemble beats the raw sample on most cells, with the largest gains at the smallest budgets and on the rich-response datasets (Figure 3d–g); the win over LPMC is modest but stays positive in the mean across datasets, cohorts, coverage, and query depth (Figure 4e–h).

Both applications replicate on the Every Eval Ever suite. The second suite sharpens the completion contrast (Table 1; lever sweeps in Appendix B): at $m=1$ PKPS attains a third of the sample’s error (0.076 vs. 0.224), and on missing cells the ensemble beats LPMC at *every* operating point, including full cohort (0.069 vs. 0.096 at $n=45$)—this higher-rank matrix favors the response channel. Table 1 collects both applications on both suites; the ensemble is best or tied nearly everywhere.

The results are insensitive to representation hyperparameters. Varying the MDS dimension over 4–24 moves PKPS’s MAE by less than 0.005, and the cross-validated bandwidth matches the best fixed choice (Appendix C).

5 Discussion

Comparing models does not require responses to identical queries; responses to similar queries suffice. Evaluation data is rarely paired in practice: each model answers the queries its evaluators chose, and methods that require a shared block of identical queries do not apply. PKPS removes this requirement by weighting response comparisons with a query kernel, recovering paired DKPS in the $\sigma \rightarrow 0$ limit. Our central finding is that benchmark prediction remains possible under unpaired evaluation: as $\rho \rightarrow 0$, DKPS and IRT degrade sharply while PKPS does not. This yields two applications from a single embedding, denoising cheap evaluations and predicting never-run cells, with the ensemble performing at least as well as either channel alone.

Limitations and future work. PKPS embeds every observed response and query (embedder effects not ablated); the block-diagonal instantiation disables cross-domain bridging (a shared response space is untested); the affinity is quadratic in queries per model pair. Theorem 1 is existential (no rate in n) and the strict DKPS separation remains a conjecture. Finally, per-cell wins over score-only baselines are modest on dense low-rank matrices and decisive in the sparse, noisy, unpaired regime that motivates the problem.

Reproducibility statement

All methods are released as a documented Python package (the PKPS estimator, matrix completion, and IRT baselines), with the data-preparation, experiment, and figure scripts for both the synthetic study and the HELM suite; every experiment is seeded, all real-data numbers are means over 16 seeds (synthetic: 50), and hyperparameters are selected by the leak-free cross-validation described in Section 3 (leakage is checked by an automated test that corrupts one model’s held-out scores and verifies its predictions are unchanged). Appendix A lists the suite composition, embedding models, and all hyperparameter settings.

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A Experimental details

Suite. 18 tasks over four datasets from HELM [Liang et al., 2023]: MATH (7 subjects), WMT-14 (5 translation directions), MedQA, and LegalBench (5 subsets), on the 93 models evaluated on all four; Table 2 lists every task with its query count (the loader subsamples each task to at most 120 queries with a fixed seed, so datasets contribute comparably). MATH and WMT-14 responses are free text and are embedded with gemini-embedding-001 (3072-d); MedQA and LegalBench responses are single-token answers, encoded one-hot. Queries share a single gemini-embedding-001 embedding space. Response and query embeddings are reduced by PCA fit on observed rows only, with dimension chosen by the second Zhu–Ghodsai elbow.

Table 2: HELM suite composition: 18 tasks over four datasets, with the number of unique queries per task. Every task is scored for all 93 shared models; the loader caps each task at 120 queries (seeded subsample).

Dataset	Task	Queries	Dataset	Task	Queries
MATH	algebra	135	WMT-14	cs-en	873
MATH	counting & probability	39	WMT-14	de-en	776
MATH	geometry	38	WMT-14	fr-en	754
MATH	intermediate algebra	52	WMT-14	hi-en	387
MATH	number theory	30	WMT-14	ru-en	613
MATH	prealgebra	86	MedQA	med.qa	1000
MATH	precalculus	57	LegalBench	abercrombie	95
LegalBench	corporate lobbying	490	LegalBench	citizenship questions	1000
LegalBench	function of decision section	367	LegalBench	proa	95

Model coverage (HELM). The 93 shared models span 20 developers: 01-ai-yi-34b, 01-ai-yi-6b, 01-ai-yi-large-preview, AlephAlpha.luminous-base, AlephAlpha.luminous-extended, AlephAlpha.luminous-supreme, ai21-j2-grande, ai21-j2-jumbo, ai21-jamba-1.5-large, ai21-jamba-1.5-mini, ai21-jamba-instruct, allenai-olmo-7b, amazon-nova-lite-v1, amazon-nova-micro-v1, amazon-nova-pro-v1, anthropic-claude-2.0, anthropic-claude-2.1, anthropic-claude-3-5-haiku-20241022, anthropic-claude-3-5-sonnet-20240620, anthropic-claude-3-5-sonnet-20241022, anthropic-claude-3-haiku-20240307, anthropic-claude-3-opus-20240229, anthropic-claude-3-sonnet-20240229, anthropic-claude-instant-1.2, anthropic-claude-instant-v1, anthropic-claude-v1.3, cohere-command, cohere-command-light, cohere-command-r, cohere-command-r-plus, databricks-dbrx-instruct, deepseek-ai-deepseek-llm-67b-chat, deepseek-ai-deepseek-v3, google-gemini-1.0-pro-001, google-gemini-1.0-pro-002, google-gemini-1.5-flash-001, google-gemini-1.5-flash-002, google-gemini-1.5-pro-001, google-gemini-1.5-pro-002, google-gemini-1.5-pro-preview-0409, google-gemini-2.0-flash-exp, google-gemma-2-27b-it, google-gemma-2-9b-it, google-gemma-7b, google-text-bison@001, google-text-unicorn@001, meta-llama-2-13b, meta-llama-2-70b, meta-llama-2-7b, meta-llama-3-70b, meta-llama-3-8b, meta-llama-3.1-405b-instruct-turbo, meta-llama-3.1-70b-instruct-turbo, meta-llama-3.1-8b-instruct-turbo, meta-llama-3.2-11b-vision-instruct-turbo, meta-llama-3.2-90b-vision-instruct-turbo, meta-llama-3.3-70b-instruct-turbo, meta-llama-65b, microsoft-phi-2, microsoft-phi-3-medium-4k-instruct, microsoft-phi-3-small-8k-instruct, mistralai-mistral-7b-instruct-v0.3, mistralai-mistral-7b-v0.1, mistralai-mistral-large-2407, mistralai-mistral-medium-2312, mistralai-mixtral-8x22b, mistralai-mixtral-8x7b-32kseqn, mistralai-open-mistral-nemo-2407, nvidia-nemotron-4-340b-instruct, openai-gpt-3.5-turbo-0613, openai-gpt-4-0613, openai-gpt-4-1106-preview, openai-gpt-4-turbo-2024-04-09, openai-gpt-4o-2024-05-13, openai-gpt-4o-2024-08-06, openai-gpt-4o-mini-2024-07-18, openai-text-davinci-002, openai-text-davinci-003, qwen-qwen1.5-110b-chat, qwen-qwen1.5-14b, qwen-qwen1.5-32b, qwen-qwen1.5-72b, qwen-qwen1.5-7b, qwen-qwen2-72b-instruct, qwen-qwen2.5-72b-instruct-turbo, qwen-qwen2.5-7b-instruct-turbo, snowflake-snowflake-arctic-instruct, tiuae-falcon-40b, tiuae-falcon-7b, upstage-solar-pro-241126, writer-palmyra-x-004, writer-palmyra-x-v2, writer-palmyra-x-v3.

PKPS. RBF query kernel with bandwidth selected per run by leave-one-model-out k -NN ($k=8$) on the observed scores over the grid $\sigma \in \{0.03, 0.1, 0.3, 1, 3\} \times \text{median pairwise query distance}$, plus the near-delta (DKPS) limit; linear response kernel on real data, RBF on synthetic. Response blocks are capped at 48 PCA components per dataset and the shared query space at 64; each (model, query) pair uses one sampled response ($r=1$). Models are embedded by classical MDS with $d=8$ throughout, following Helm et al. [2026] and fixed a priori ($d = \text{latent dimension on synthetic}$). Appendix C shows the results are insensitive to d over 4–24, and that the cross-validated bandwidth tracks the best fixed choice.

Prediction head. All embedding methods share one head: logit-transformed scores, per-task standardization to unit variance, a two-way (model + task) additive bias, and a distance-weighted k -NN ($k=5$) regression of the residual on the model embedding, fit *leave-one-family-out*: all models from the target’s developer are excluded from its training set, so predictions never lean on near-duplicate siblings [Helm et al., 2026]. Held-out cells are predicted directly; observed cells leave their own target out.

Baselines. Low-rank matrix completion (LRMC) follows BenchPress [Zeng and Papailiopoulos, 2026]: logit space, per-task standardization, two-way bias, rank-2 ALS with ridge $\lambda=0.1$, averaged over 5 random initializations; on observed cells it is cross-fit so a cell’s prediction never sees its own sample. IRT is a 1PL (Rasch) model on binary tasks: item difficulties by maximum likelihood (L-BFGS) on the largest fully-observed (model $\times$ item) block, per-model ability by bounded scalar MLE, clipped to $[-4, 4]$. The sample baseline is the cell’s own m -query mean.

Ensemble. On observed cells, the sample mean is shrunk toward the PKPS estimate with per-cell precision weight $e_{PKPS}^2 / (e_{PKPS}^2 + \hat{p}(1 - \hat{p})/m)$ on the sample, where e_{PKPS}^2 is the embedding’s depth-weighted disagreement with the observed samples of that model; on missing cells, PKPS and LRMC

are blended with a weight fit by held-out cross-validation against observed scores. No quantity ever uses the held-out full-eval scores.

Sweeps and reporting. All real-data sweeps use 16 seeds; curves and table entries report the per-seed mean over tasks, then the mean (± 1 SEM) over seeds. Query efficiency: budgets $m \in \{1, 2, 4, 8, 16, 32\}$ at full cohort and coverage; cohort $n \in \{10, 20, 40, 60, 93\}$ (HELM) or $\{10, 20, 30, 45\}$ (EEE) and task coverage $p_{\text{task}} \in \{0.2, 0.35, 0.5, 0.7, 0.9\}$, both at the few-query point $m=2$. Completion: task coverage as above and query depth $p_{\text{query}} \in \{0.1, 0.25, 0.5, 0.75, 1.0\}$ with cohort lines $n \in \{10, \text{full}\}$; task count $T \in \{4, 8, 12, 18\}$ (HELM) or $\{4, 8, 12, 16\}$ (EEE); the cohort sweep holds $p_{\text{task}}=p_{\text{query}}=0.5$.

Synthetic study. 50 seeds per configuration; latent dimension 5, observed embedding dimension 20, score noise 0.1, response noise 0.5; both kernels RBF; k -NN regression with $k=5$. Panel-specific settings are given in each figure title.

Compute. All experiments run on a single multi-core CPU node; text embeddings are precomputed once through the embedding API and cached. Embedding the EEE suite (30,304 pooled responses plus 7,665 queries, $\sim 11\text{M}$ tokens) took ~ 13 minutes and under \$2 of API credit; each 16-seed sweep completes in minutes (EEE) to hours (HELM) on the same node.

B The Every Eval Ever suite

The second suite is built from the Every Eval Ever datastore [EvalEval Coalition, 2026], an MIT-licensed, community-maintained collection of item-level evaluation results in a unified format (inputs, raw model outputs, and per-item scores). We take the five benchmarks with the widest item-level model coverage—MATH-MC, GSM-MC, GPQA-Diamond, JudgeBench, and RewardBench 2—restricted to the 45 models present on all five; sub-tasks (MATH-MC levels 1–5; four JudgeBench and five RewardBench 2 categories) give 16 tasks (Table 3). Every domain is free text (including the judging benchmarks’ rationales), so all responses and queries are embedded with gemini-embedding-001 and blocked by benchmark exactly as in Appendix A. Each cell’s observable pool is 32 queries drawn with a model-dependent seed, so two models share only incidental queries ($\sim 10\%$); the query-efficiency budget m and the completion depth p_{query} subsample this pool, and the prediction target is always the cell’s full-depth mean over all items, which is never embedded. All scores are binary, so IRT applies to every task (on the HELM suite it excludes WMT). Figures 5 and 6 repeat the lever sweeps of Figures 3 and 4 on this suite; Table 1 summarizes.

Table 3: EEE suite composition: 16 tasks over five benchmarks, with the number of items per (model, task) cell. Every task is fully answered by all 45 shared models; each cell’s observable pool is 32 items, sampled per model.

Benchmark	Task	Items	Benchmark	Task	Items
MATH-MC	level 1	430	JudgeBench	coding	42
MATH-MC	level 2	882	JudgeBench	knowledge	154
MATH-MC	level 3	1115	JudgeBench	math	56
MATH-MC	level 4	1199	JudgeBench	reasoning	98
MATH-MC	level 5	1288	RewardBench 2	factuality	475
GSM-MC	gsm_mc	1319	RewardBench 2	focus	495
GPQA-Diamond	gpqa_diamond	198	RewardBench 2	math	183
RewardBench 2	precise if	160	RewardBench 2	safety	450

Model coverage (EEE). The 45 models shared by all five benchmarks span 13 developers: cohere/c4ai-command-a-03-2025, cohere/c4ai-command-r-08-2024,

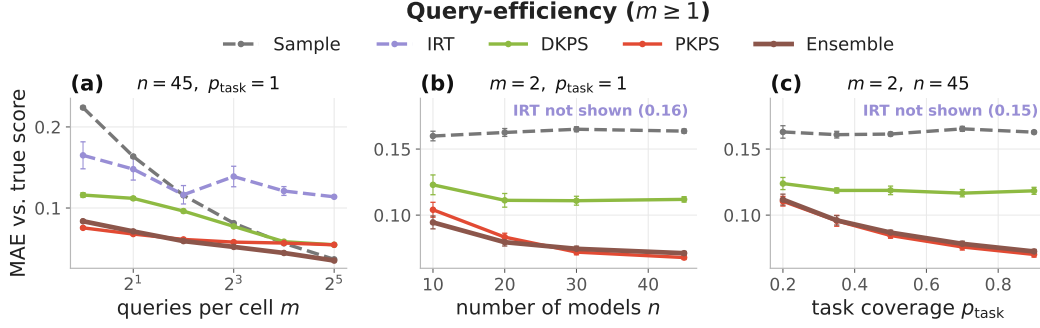


Figure 5: **Query efficiency on the Every Eval Ever suite** (45 models, 16 tasks; MAE vs. full-depth score, ± 1 SEM over 16 seeds). Same protocol as Figure 3. PKPS at $m=1$ attains a third of the sample’s error and the sample needs $m \approx 8$ to catch up; DKPS trails because pools are unpaired, and IRT is poorly identified. The ensemble is best from $m \geq 2$ on.

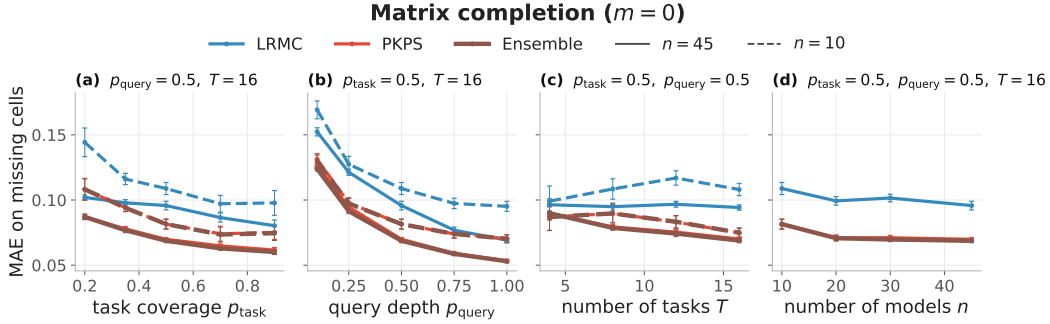


Figure 6: **Matrix completion on the Every Eval Ever suite** (MAE on missing cells, ± 1 SEM; solid $n=45$, dashed $n=10$). Same protocol as Figure 4. Unlike the HELM suite, the embedding and ensemble beat LRM at every operating point—including full cohort and coverage—consistent with the higher effective rank of this modern score matrix.

cohere/c4ai-command-r-plus-08-2024, cohere/c4ai-command-r7b-12-2024, cohere/command-a-reasoning-08-2025, deepseek/deepseek-r1-0528, deepseek/deepseek-v3-1-terminus, deepseek/deepseek-v3-2, deepseek/deepseek-v3-2-speciale, deepseek/deepseek-v4-flash-fp8, deepseek/deepseek-v4-pro, google/gemini-3-1-pro-preview, google/gemma-2-27b-it, google/gemma-2-9b-it, google/gemma-3-12b-it, google/gemma-3-27b-it, google/gemma-4-e2b-it, google/gemma-4-e4b-it, llm360/k2-v2-instruct, meta/llama-4-maverick-17b-128e-instruct-fp8, meta/meta-llama-3-1-8b-instruct, minimax/minimax-m2-5, mistral/mistral-small-4-119b-2603, moonshot/kimi-k2-5, moonshot/kimi-k2-6, moonshot/kimi-k2-thinking, openai/gpt-5-5, openai/gpt-5-mini, openai/gpt-5-nano, openai/gpt-oss-120b, openai/gpt-oss-20b, qwen/qwen3-30b-a3b-thinking-2507, qwen/qwen3-5-122b-a10b, qwen/qwen3-5-2b, qwen/qwen3-5-397b-a17b, qwen/qwen3-6-35b-a3b, qwen/qwen3-6-plus, qwen/qwen3-next-80b-a3b-thinking, stepfun/step-3-5-flash, xiaomi/mimo-v2-flash, zai-org/glm-4-6-fp8, zai-org/glm-4-7-flash, zai-org/glm-4-7-fp8, zai-org/glm-5-1-fp8, zai-org/glm-5-fp8.

C Sensitivity analyses

Figure 7 probes the two representation hyperparameters on the query-efficiency experiment (16 seeds throughout). **MDS dimension (a, b)**. Repeating the budget sweep at $d \in \{4, 8, 24\}$ moves PKPS’s MAE by less than 0.005 at every budget on both suites (at the smallest budgets, smaller d is very slightly better—fewer coordinates to estimate from one noisy query per cell); the paper’s choice ($d=8$, following Helm et al., 2026) is never the worst point at any budget. **Query bandwidth (c)**. Fixing the bandwidth at each grid multiplier instead of cross-validating traces the expected curve:

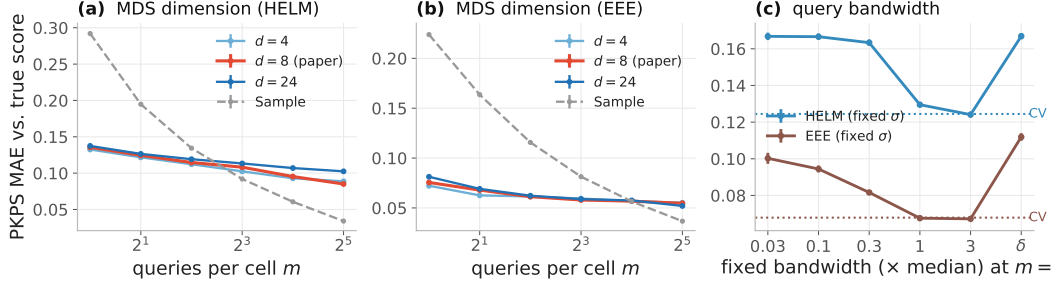


Figure 7: **Sensitivity analyses** on the query-efficiency experiment (± 1 SEM, 16 seeds). **(a, b)** PKPS MAE vs. budget at MDS dimension $d \in \{4, 8, 24\}$ (paper choice in red; raw sample dashed): the curves are nearly indistinguishable on both suites. **(c)** MAE at $m=2$ with the query bandwidth *fixed* at each grid multiplier ($\delta = \text{exact matching}$); dotted lines mark the cross-validated selection, which matches the best fixed choice on both suites.

near-delta multipliers and the exact-matching δ limit are substantially worse (unpaired pools leave them little to compare), broad multipliers are best, and the leak-free cross-validated choice matches the best fixed multiplier on both suites (0.126 vs. 0.125 on HELM, 0.069 vs. 0.068 on EEE)—the one hyperparameter that matters is the one the method selects automatically. Sensitivity to the number of reference models is part of the main experiments (the cohort sweeps of Figures 3b and 4d).

D Proof of Theorem 1

Population quantities. Queries for model a are drawn i.i.d. from a design distribution over embedded queries; write $z = (u, x_a(u))$ for an embedded query–response pair (response noise included) and P_a for its law. By Assumption 3 the kernels are bounded: $k_Q \in [0, 1]$ after normalization (RBF) and $|k_R| \leq B_R$. Define the population affinity

$$A_{ab}^* = \frac{\mathbb{E}[k_Q(u, u') k_R(x_a(u), x_b(u'))]}{\mathbb{E}[k_Q(u, u')]}, \quad z \sim P_a, z' \sim P_b \text{ independent},$$

population squared distances $D^{*2}(a, b) = A_{aa}^* + A_{bb}^* - 2A_{ab}^*$, the centered Gram matrix $B^* = -\frac{1}{2}JD^{*2}J$ with $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, and population perspectives $\Psi^* \in \mathbb{R}^{n \times d}$ its rank- d spectral embedding, which exists with $\lambda_d(B^*) > 0$ by Assumption 3 (cf. Definition 1).

Lemma 1 (Affinity concentration). *Under Assumption 3, for models a, b with $\min(|Q_a|, |Q_b|) \geq m$ and any $\delta \in (0, 1)$: with probability at least $1 - \delta$,*

$$|A_{ab} - A_{ab}^*| \leq \frac{2(1+B_R)}{\kappa^2} t_m(\delta), \quad t_m(\delta) = \bar{B} \sqrt{\frac{\log(4/\delta)}{2m}}, \quad \bar{B} = \max(2B_R, 1),$$

provided $t_m(\delta) \leq \kappa/2$.

Proof. The numerator $N = \frac{1}{|Q_a||Q_b|} \sum_{j,l} k_Q(u_j, u'_l) k_R(x_{aj}, x_{bl})$ and denominator $Z = \frac{1}{|Q_a||Q_b|} \sum_{j,l} k_Q(u_j, u'_l)$ of Eq. 2 are two-sample U -statistics of order (1,1) in the i.i.d. pairs $z_j \sim P_a, z'_l \sim P_b$, with kernels of range at most $2B_R$ and 1 respectively (response noise enters only through the bounded kernel argument, so the number of responses per query may stay fixed). By Hoeffding’s inequality for two-sample U -statistics [Hoeffding, 1963, Section 5], each deviates from its mean by more than its range times $\sqrt{\log(4/\delta)/2m}$ with probability at most $\delta/2$; since $\bar{B}$ dominates both ranges, a union bound gives $|N - N^*| \leq t_m(\delta)$ and $|Z - Z^*| \leq t_m(\delta)$ jointly with

probability $\geq 1 - \delta$. On this event, since $Z^* \geq \kappa$ by Assumption 3 and $t_m(\delta) \leq \kappa/2$, we have $Z \geq \kappa/2$, and

$$|A_{ab} - A_{ab}^*| = \frac{|NZ^* - N^*Z|}{ZZ^*} \leq \frac{|N - N^*|}{Z} + \frac{|N^*||Z - Z^*|}{ZZ^*} \leq \frac{2t_m(\delta)}{\kappa} + \frac{2B_R t_m(\delta)}{\kappa^2} \leq \frac{2(1 + B_R)}{\kappa^2} t_m(\delta),$$

using $|N^*| \leq B_R$ and $\kappa \leq 1$.

□

Lemma 2 (Perspective perturbation). *Let $\varepsilon_\infty = \max_{a,b} |A_{ab} - A_{ab}^*|$. There is an orthogonal $W \in \mathbb{R}^{d \times d}$ such that*

$$\max_i \|\hat{\psi}_i - W\psi_i\| \leq \frac{25\sqrt{d} n \varepsilon_\infty}{\sqrt{\lambda_d(B^*)}}.$$

Proof. Entrywise, $|D^2 - D^{*2}| \leq 4\varepsilon_\infty$, and double centering inflates an entrywise bound by at most a factor of 4, so $E = B - B^*$ satisfies $\|E\|_{\text{op}} \leq \|E\|_F \leq 8n\varepsilon_\infty$. Let B_d be the top- d truncation of B that classical MDS embeds ($B_d = \hat{\Psi}\hat{\Psi}^\top$, positive semidefinite). By Weyl's inequality every eigenvalue of B beyond the d -th, and every negative eigenvalue, has magnitude at most $\|E\|_{\text{op}}$ (since $\lambda_{d+1}(B^*) = 0$ and $B^* \succeq 0$), so $\|B_d - B\|_{\text{op}} \leq \|E\|_{\text{op}}$. The matrix $B_d - B^*$ has rank at most $2d$, hence

$$\|B_d - B^*\|_F \leq \sqrt{2d} \|B_d - B^*\|_{\text{op}} \leq \sqrt{2d} (\|B_d - B\|_{\text{op}} + \|E\|_{\text{op}}) \leq 2\sqrt{2d} \|E\|_F.$$

Both $B_d = \hat{\Psi}\hat{\Psi}^\top$ and $B^* = \Psi^*\Psi^{*\top}$ are positive semidefinite of rank at most d , so by the Procrustes alignment lemma of Tu et al. [2016, Lemma 5.4],

$$\min_{W \in O(d)} \|\hat{\Psi} - \Psi^*W\|_F^2 \leq \frac{\|B_d - B^*\|_F^2}{2(\sqrt{2} - 1)\lambda_d(B^*)}.$$

Combining, $\max_i \|\hat{\psi}_i - W\psi_i\| \leq \|\hat{\Psi} - \Psi^*W\|_F \leq \frac{2\sqrt{2d}}{\sqrt{2(\sqrt{2}-1)}} \cdot \frac{8n\varepsilon_\infty}{\sqrt{\lambda_d(B^*)}} \leq \frac{25\sqrt{d} n \varepsilon_\infty}{\sqrt{\lambda_d(B^*)}}$. Nearest-neighbor distances are invariant to W . □

Proof of Theorem 1. Take scores in $[0, 1]$ without loss of generality (Assumption 1). Fix $\varepsilon > 0$ and set targets $c \leq \sqrt{\varepsilon/2}/(8\gamma)$ for the estimation error and $\varepsilon' \leq \sqrt{\varepsilon/2}/(2\gamma)$ for the population nearest-neighbor distance. By Assumption 2 there is $n_0(\varepsilon')$ such that for $n \geq n_0$, a reference model within population distance ε' of the target exists except with probability at most $\varepsilon/8$. Fix such an n . Applying Lemma 1 with confidence $\delta = \varepsilon/(8n^2)$ to every pair, a union bound and Lemma 2 give $\max_i \|\hat{\psi}_i - W\psi_i\| \leq c$ except with probability at most $\varepsilon/8$, once

$$m \geq C' \frac{d(1 + B_R)^2 \bar{B}^2 n^2 \log(n/\varepsilon)}{\kappa^4 \lambda_d c^2} = \text{poly}(n, 1/\kappa, 1/\varepsilon),$$

for an absolute constant C' (this choice also satisfies the proviso $t_m \leq \kappa/2$ of Lemma 1), and does not involve the pairing fraction ρ . On the intersection of the two events, for a target model f with nearest estimated reference neighbor f^* (perspectives $\psi^*, \hat{\psi}^*$; the target's $\psi, \hat{\psi}$),

$$|\hat{y}(f) - y(f)| \leq \gamma \|\psi^* - \psi\| \leq \gamma (\|\hat{\psi}^* - W\psi^*\| + \|\hat{\psi}^* - \hat{\psi}\| + \|\hat{\psi} - W\psi\|) \leq \gamma(2c + \hat{\delta}),$$

using Assumption 1 and orthogonal invariance. The estimated nearest-neighbor distance $\hat{\delta}$ is at most the estimated distance to the ε' -close reference, itself at most $\varepsilon' + 2c$ by the same alignment argument, so $|\hat{y} - y| \leq \gamma(4c + \varepsilon') \leq \sqrt{\varepsilon/2}$. Squaring, and adding the two failure events (squared errors are bounded by 1), $\text{MSE}(\hat{y}) \leq \varepsilon/2 + \varepsilon/4 \leq \varepsilon$. □

Relative to the sample score. The sample score at budget m is the mean of m i.i.d. per-query scores and uses no reference models, so its MSE is σ_q^2/m for every n , where σ_q^2 is the per-query score variance ($\leq 1/4$ for scores in $[0, 1]$). For any $\varepsilon < \sigma_q^2/m$ at the budget of Theorem 1, the theorem’s estimator satisfies $\text{MSE} \leq \varepsilon < \sigma_q^2/m$ for all sufficiently large n : query efficiency relative to the sample score in the sense of Definition 2.

Contrast with paired DKPS. With $k_Q = \delta$, only identically-shared query pairs enter the double sum, so the U -statistic runs over the ρM_{ij} shared queries and the rate in Lemma 1 becomes $O_P((\rho M_{ij})^{-1/2})$: DKPS’s guarantee requires $\rho M_{ij} \rightarrow \infty$ and vacates in the unpaired limit $\rho \rightarrow 0$, where its distance is undefined (Section 3). This asymmetry motivates Conjecture 1; a formal lower bound for DKPS is future work.